

YIWEN (CINDY) MA

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EDUCATION

University of Bristol *MSc Business Analytics*

Sep 2025 - Jan 2027

Relevant Modules: Business Data Analytics, Behaviour, Decision Making and Analytics, Optimisation, Data Analytics.

Nanjing Audit University *BSc Finance*

Sep 2021 - Jun 2025

Overall Average: 83% | **GPA:** 3.19/4.00 | **Top** 20%

Relevant Modules: Real Estate Finance, Management of Commercial Banks, Microeconometrics, Corporate Finance, Econometrics,

PROFESSIONAL EXPERIENCE

IBM | SkillsBuild Learning Pathway Recommendation System

Jan 2026 - May 2026

Data Analytics Intern | London, UK

- Designed and distributed a user research survey on SkillsBuild course matching needs, collecting **38 valid responses** to inform the recommendation criteria and weighting.
- Independently developed a rule-based recommendation engine using **six dimensions**: role fit, degree relevance, skill gaps, difficulty level, prerequisites and interest keywords. **Scored 230+ courses**, normalised results into a 65–99% match range, and generated tiered learning pathways from beginner to advanced.
- Conducted user testing with **65 participants**, **89% of whom rated the recommendations 4–5 stars**. Implemented the scoring logic in the front end, worked with a six-person international team to deliver the IBM client project, launched the final web application on Firebase, and documented the system architecture, data flow and recommendation logic.

ByteDance

Jun 2025 - Aug 2025

Product Management Intern | Beijing, China

- Conducted semi-structured interviews with three professional comic studios and mapped real production workflows, translating user feedback into four requirement areas: line quality, colouring accuracy, style consistency and workflow efficiency; prioritised issues by frequency, impact on output quality and implementation difficulty to support iteration of AI-assisted line-art and colouring features.
- Designed a 12-metric model evaluation framework covering line completeness**, colour-region accuracy, colour overflow and style consistency; built a fixed test set spanning different character counts, line complexity, occlusion levels and colour regions, and compared eight model versions to support model selection and parameter tuning.
- Built a pilot-user monitoring dashboard tracking activity, task completion, feature usage and retention, identifying first-use drop-off, low engagement with the AI colouring entry point and repeated rework; subsequent iterations were associated **with a 20% relative increase in user activity, an 8% relative increase in AI-colouring entry clicks and a 7% relative increase in pilot-user retention**.

Shanxi Securities Co., Ltd., Industry Research Institute

Jan 2025 - May 2025

Research Intern | Shanghai, China

- Conducted industry and competitor research across data-centre power equipment and humanoid robotics, comparing market size, cost structures, supply chains, technology routes and **capacity plans across 30+ companies; contributed to 5 sector and strategy reports**.
- Built and analysed structured datasets **using Excel, Python and SQL, including a 2020–2024 urban rail procurement database covering 217 projects, 26,088 vehicles** and approximately RMB 206.3bn in tender value, supporting market sizing and demand analysis.
- Researched 18 global humanoid robotics companies**, comparing supply-chain positioning, technology routes, production plans, target pricing and key component specifications; contributed to the core research materials for the annual robotics strategy report.

Grant Thornton China

Nov 2024 - Jan 2025

Consulting Intern | Shenzhen, China

- Analysed more than 1,000 transaction records per day, identified 38 anomalies** and structured results in Excel-based issue logs to support risk reviews and management decision-making.
- Built NPV models, conducted sensitivity and budget-variance analyses, and translated financial and operational data into client-facing advisory findings.

PROJECT EXPERIENCE

CompositeLens-AI Industrial Inspection Product

Aug 2026

Winner, AI Hackathon Advanced Manufacturing Challenge

- Productised the team's defect segmentation model into a browser-based inspection tool supporting single-image, batch and folder uploads. Integrated server-side inference and automated PDF reporting with defect measurements and decision rationale, allowing users to run inspections without local setup or GPU access.
- Designed a human-in-the-loop review mechanism using decision-margin distance, multi-view consistency, image quality and physical-scale checks. Achieved an 84.6% automation rate and 95.7% accuracy with zero missed defects across 247 validation samples, while routing 40% of model errors to manual review.
- Standardised evaluation across eight modelling approaches, with the final solution achieving zero missed defects and a 0.9949 validation score on 268 independent samples. Built the pricing and ROI model, estimating approximately £25k in annual customer value and proposing a £6k annual licence, with an estimated payback period of around three months.

Humanoid Robotics Industry Research & Automated Tracking System [Project Link](#)

Aug 2026 - Sep 2026

- Built a humanoid robotics research platform from scratch, comparing market forecasts from seven institutions to identify differences in assumptions and measurement methods. Tracked technology roadmaps, production plans and target pricing across 10 global companies, alongside eight private funding rounds and major IPO developments.
- Built a multi-agent cross-validation workflow to trace and verify 87 data points and references, automatically identifying and correcting three data issues while keeping research conclusions traceable to primary sources.
- Designed four parallel Agent workflows to automate research, verification, content generation and publishing through scheduled cloud agents and the GitHub API. Reduced a full update cycle to around 90 seconds, ran continuously for 10 days, automatically skipped publishing when no new information was found, and supported bilingual Chinese and English output.

Global Semiconductor Peer Benchmarking Study

May 2026 - Aug 2026

- Built a FY2020–2024 comparable panel using Compustat/WRDS and Refinitiv Eikon, selecting 40 pure-play semiconductor companies from 58 candidates across IC design, foundry and OSAT segments. Standardised currencies, fiscal years and price levels to create 200 firm-year observations, with 11 IDMs retained as a separate reference group.
- Found that IC design firms had an average efficiency score of 0.569, significantly higher than 0.304 for foundries ($p = 0.002$), while differences across Taiwan, the US and China were not statistically significant, suggesting that business model explained more of the efficiency gap than geography.
- Applied bootstrapped truncated regression to identify efficiency drivers, finding a robust negative association between foundry status and efficiency of approximately 40 percentage points. Peer-group composition also materially affected rankings, with median efficiency increasing 32% when the comparison set was reduced to 11 firms.

Lululemon Social Media & Customer Insights Analysis

Mar 2026 - Apr 2026

- Collected and cleaned 2,180 Reddit and YouTube comments using Python, Reddit's public JSON interface and Selenium. Capped comments per post to reduce sampling bias, built a six-category keyword framework, and applied VADER sentiment analysis and LDA topic modelling to identify recurring customer concerns.
- Found that 69.2% of comments were positive, 18.0% neutral and 12.8% negative. Negative sentiment reached 20% for price and perceived value and 17% for durability, while sizing accounted for 19% of all discussions. Translated these findings into recommendations for sizing guidance, material communication and value messaging.

Marketing Budget Optimisation Model

Oct 2025 - Dec 2025

- Analysed 40,000+ campaign records and ranked campaigns by conversion efficiency, then built a Python linear programming model to optimise budget allocation while maintaining conversion, impression and engagement targets.
- The model projected a 25% reduction in marketing spend against a baseline average CPA of US\$635 by reallocating budget from lower-performing campaigns to more efficient ones.

SKILLS & INTERESTS

- **Programming:** Python (Pandas, NumPy, scikit-learn, statsmodels and Selenium), SQL, JavaScript.
- **Data Analysis:** Data cleaning, exploratory data analysis, regression, sentiment analysis, topic modelling, machine learning, optimisation and cross-validation.
- **Visualisation:** Tableau, Matplotlib, Power BI.
- **Financial Analysis:** NPV modelling, sensitivity analysis, financial forecasting, financial statement analysis, comparable-company benchmarking, valuation multiples and sector research.
- **Certifications:** Harvard University edX CS109x Verified Certificate; Deloitte Australia Data Analytics Job Simulation, Forage.
- **Languages:** Mandarin Chinese (Native), English (IELTS 6.5).
- **Interests:** Guzheng, Photography (Adobe Lightroom, Capture One).